Laporan Pemrograman Mobile 4



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2026

Pertanyaan Refleksi :

1. Apa yang terjadi bila Future dibuat di dalam build() dan bukan di initState()? Mengapa?
2. Apa perbedaan snapshot.hasError dengan memeriksa response.statusCode? Mengapa keduanya diperlukan?
3. Mengapa data JSON sebaiknya diubah menjadi class model, bukan dipakai langsung sebagai Map?
4. Mengapa UI wajib menyediakan status loading dan error, bukan hanya status data?

Jawab :

1. Http request akan dikirim ulang secara terus menerus ketika widget melakukan rebuild, hal ini terjadi karena future harusnya dipanggil sekali sedangkan build() akan memanggil berulang kali karena itulah biasa future diletakan di initstate().
2. response.statuscode digunakan http untuk mengecek balasan server, sementara snapshot.haserror digunakan untuk mendeteksi apakah proses future melempar exception. Keduanya diperlukan karena response.statuscode berguna untuk mendeteksi kegagalan dari server dan snapshot.haserror berfungsi menangkap exception agar UI dapat menampilkan galat.
3. JSON diubah menjadi class model untuk mengurangi kesalahan typing, mempermudah proses perubahan struktur kode, dan struktur data yang lebih teratur.
4. status loading wajib ada supaya pengguna sadar bahwa program sedang loading dan status error juga wajib ada supaya pengguna tahu apa yang salah dan bisa menyelesaikannya.

Bagian A :

```dart
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
 const MyApp({super.key});

 @override
 Widget build(BuildContext context) {
   return MaterialApp(
     title: 'Praktikum 4',
     theme: ThemeData(
       colorSchemeSeed: Colors.blue,
       useMaterial3: true,
     ),
     home: const SalamPage(),
   );
 }
}

Future<String> ambilSalam() async {
 await Future.delayed(const Duration(seconds: 2));
 return 'Halo dari masa depan!';
}

class SalamPage extends StatefulWidget {
 const SalamPage({super.key});

 @override
 State<SalamPage> createState() => _SalamPageState();
}

class _SalamPageState extends State<SalamPage> {
 late Future<String> _future;

 @override
 void initState() {
   super.initState();
   _future = ambilSalam();
 }

 @override
 Widget build(BuildContext context) {
   return Scaffold(
     appBar: AppBar(
       title: const Text('Demo Future'),
     ),
     body: Center(
       child: FutureBuilder<String>(
         future: _future,
         builder: (context, snapshot) {
           if (snapshot.connectionState == ConnectionState.waiting) {
```

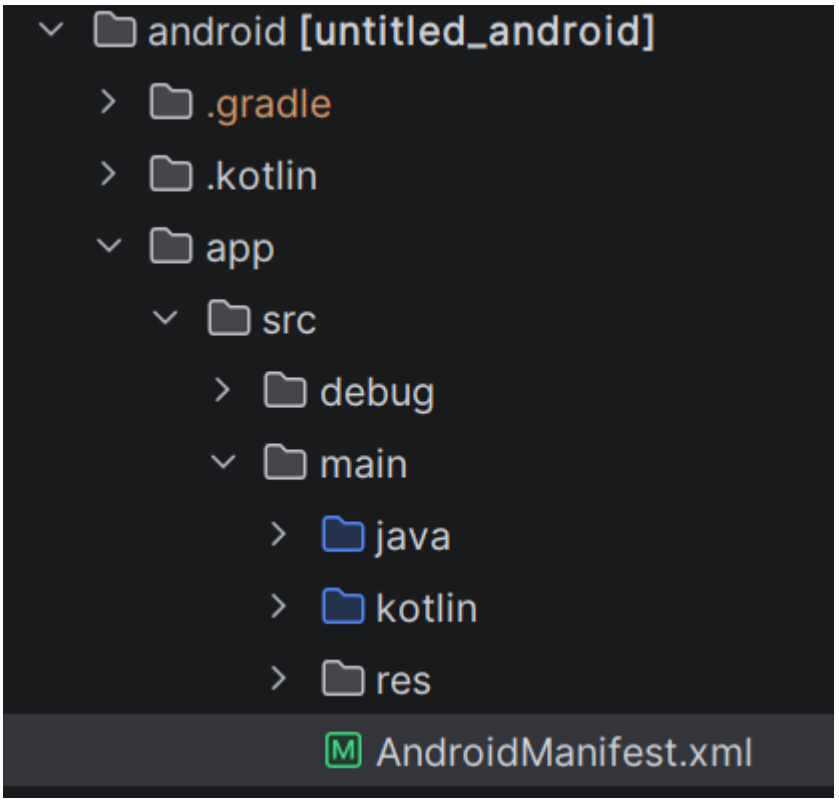
```
            return const CircularProgressIndicator();
          }
          if (snapshot.hasError) {
            return Text('Error: ${snapshot.error}');
          }
          return Text(
            snapshot.data!,
            style: const TextStyle(fontSize: 24),
          );
        },
      ),
    ),
  );
 }
}
```

Hasil :



Bagian B :

```
PS C:\Users\advan\StudioProjects\untitled> flutter pub add http
```




```
<manifest xmlns:android="http://schemas.android.com/apk/res/android">
    <uses-permission android:name="android.permission.INTERNET"/>
    <application
        android:label="untitled"
        android:name="${applicationName}"
        android:icon="@mipmap/ic_launcher">
        <activity
            android:name=".MainActivity"
            android:exported="true"
            android:launchMode="singleTop"
            android:taskAffinity=""
            android:theme="@style/LaunchTheme"

android:configChanges="orientation|keyboardHidden|keyboard|screenSize|small
estScreenSize|locale|layoutDirection|fontScale|screenLayout|density|uiMode"
            android:hardwareAccelerated="true"
            android:windowSoftInputMode="adjustResize">
            <!-- Specifies an Android theme to apply to this Activity as
soon as
                 the Android process has started. This theme is visible to
the user
                 while the Flutter UI initializes. After that, this theme
continues
                 to determine the Window background behind the Flutter UI.
-->
            <meta-data
```

```
            android:name="io.flutter.embedding.android.NormalTheme"
            android:resource="@style/NormalTheme"
            />
          <intent-filter>
              <action android:name="android.intent.action.MAIN"/>
              <category android:name="android.intent.category.LAUNCHER"/>
          </intent-filter>
      </activity>
      <!-- Don't delete the meta-data below.
           This is used by the Flutter tool to generate
GeneratedPluginRegistrant.java -->
      <meta-data
          android:name="flutterEmbedding"
          android:value="2" />
   </application>
   <!-- Required to query activities that can process text, see:
        https://developer.android.com/training/package-visibility and

https://developer.android.com/reference/android/content/Intent#ACTION_PROCE
SS_TEXT.

        In particular, this is used by the Flutter engine in
io.flutter.plugin.text.ProcessTextPlugin. -->
   <queries>
      <intent>
          <action android:name="android.intent.action.PROCESS_TEXT"/>
          <data android:mimeType="text/plain"/>
      </intent>
   </queries>
</manifest>
```

Bagian C :

```dart
import 'package:flutter/material.dart';
import 'dart:convert';
import 'package:http/http.dart' as http;

class Pengguna {
 final int id;
 final String name;
 final String email;
 final String phone;
 final String website;

 const Pengguna({
   required this.id,
   required this.name,
   required this.email,
   required this.phone,
   required this.website,
 });

 factory Pengguna.fromJson(Map<String, dynamic> json) {
   return Pengguna(
     id: json['id'] as int,
     name: json['name'] as String,
     email: json['email'] as String,
     phone: json['phone'] as String,
     website: json['website'] as String,
   );
 }
}

Future<List<Pengguna>> ambilPengguna() async {
 final uri = Uri.parse('https://jsonplaceholder.typicode.com/users');
 final response = await http.get(uri).timeout(const Duration(seconds: 10));

 if (response.statusCode != 200) {
   throw Exception('Gagal memuat data (kode ${response.statusCode})');
 }

 final List<dynamic> data = jsonDecode(response.body);
 return data
     .map((e) => Pengguna.fromJson(e as Map<String, dynamic>))
     .toList();
}

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
 const MyApp({super.key});

 @override
 Widget build(BuildContext context) {
   return MaterialApp(
```

```
      title: 'Praktikum 4',
      theme: ThemeData(
        colorSchemeSeed: Colors.blue,
        useMaterial3: true,
      ),
      home: const SalamPage(),
    );
  }
}

Future<String> ambilSalam() async {
  await Future.delayed(const Duration(seconds: 2));
  return 'Halo dari masa depan!';
}

class SalamPage extends StatefulWidget {
  const SalamPage({super.key});

  @override
  State<SalamPage> createState() => _SalamPageState();
}

class _SalamPageState extends State<SalamPage> {
  late Future<String> _future;

  @override
  void initState() {
    super.initState();
    _future = ambilSalam();
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Demo Future'),
      ),
      body: Center(
        child: FutureBuilder<String>(
          future: _future,
          builder: (context, snapshot) {
            if (snapshot.connectionState == ConnectionState.waiting) {
              return const CircularProgressIndicator();
            }
            if (snapshot.hasError) {
              return Text('Error: ${snapshot.error}');
            }
            return Text(
              snapshot.data!,
              style: const TextStyle(fontSize: 24),
            );
          },
        ),
```

```
      ),
    );
  }
}
```

Hasil :



Bagian D :

```dart
import 'dart:convert';
import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;

class Pengguna {
 final int id;
 final String name;
 final String email;
 final String phone;
 final String website;

 const Pengguna({
   required this.id,
   required this.name,
   required this.email,
   required this.phone,
   required this.website,
 });

 factory Pengguna.fromJson(Map<String, dynamic> json) {
   return Pengguna(
     id: json['id'] as int,
     name: json['name'] as String,
     email: json['email'] as String,
     phone: json['phone'] as String,
     website: json['website'] as String,
   );
 }
}

Future<List<Pengguna>> ambilPengguna() async {
 final uri = Uri.parse('https://jsonplaceholder.typicode.com/users');
 final response = await http.get(uri).timeout(const Duration(seconds: 10));

 if (response.statusCode != 200) {
   throw Exception('Gagal memuat data (kode ${response.statusCode})');
 }

 final List<dynamic> data = jsonDecode(response.body);
 return data
     .map((e) => Pengguna.fromJson(e as Map<String, dynamic>))
     .toList();
}

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
 const MyApp({super.key});

 @override
 Widget build(BuildContext context) {
   return MaterialApp(
```

```
      title: 'Praktikum 4',
      theme: ThemeData(
        colorSchemeSeed: Colors.blue,
        useMaterial3: true,
      ),
      home: const PenggunaPage(),
    );
  }
}

class PenggunaPage extends StatefulWidget {
  const PenggunaPage({super.key});

  @override
  State<PenggunaPage> createState() => _PenggunaPageState();
}

class _PenggunaPageState extends State<PenggunaPage> {
  late Future<List<Pengguna>> _future;

  @override
  void initState() {
    super.initState();
    _future = ambilPengguna();
  }

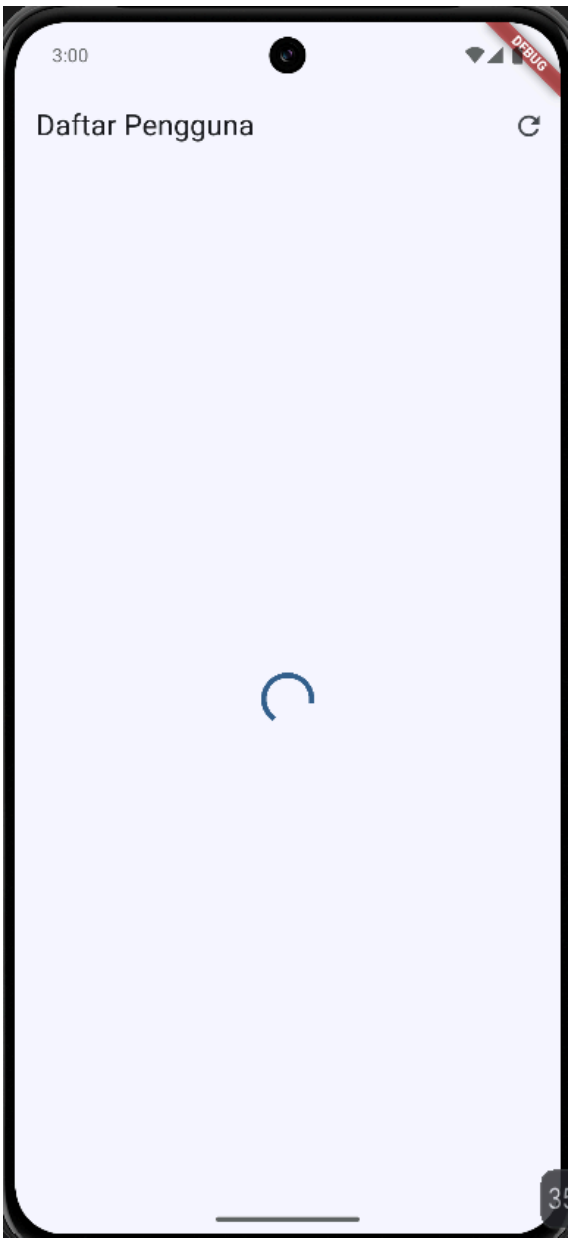
  void _muatUlang() {
    setState(() {
      _future = ambilPengguna();
    });
  }

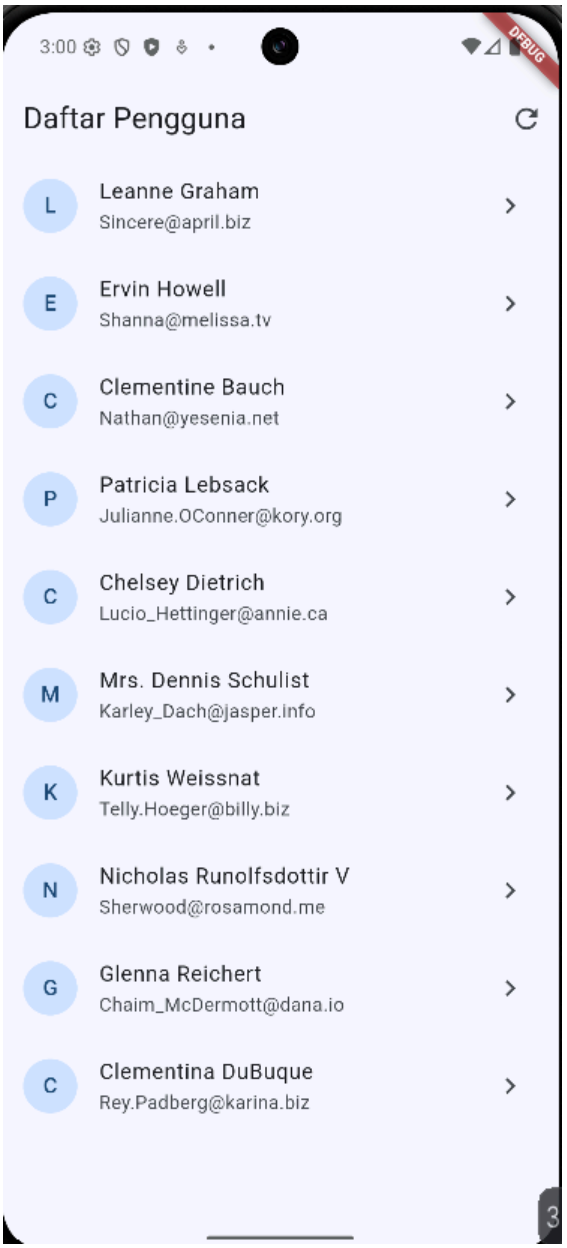
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Daftar Pengguna'),
        actions: [
          IconButton(
            icon: const Icon(Icons.refresh),
            onPressed: _muatUlang,
          ),
        ],
      ),
      body: FutureBuilder<List<Pengguna>>(
        future: _future,
        builder: (context, snapshot) {
          if (snapshot.connectionState == ConnectionState.waiting) {
            return const Center(child: CircularProgressIndicator());
          }
          if (snapshot.hasError) {
            return Center(
              child: Padding(
```

```
              padding: const EdgeInsets.all(24),
              child: Column(
                mainAxisSize: MainAxisSize.min,
                children: [
                  const Icon(
                    Icons.error_outline,
                    size: 48,
                    color: Colors.red,
                  ),
                  const SizedBox(height: 8),
                  Text(
                    'Terjadi kesalahan:\n${snapshot.error}',
                    textAlign: TextAlign.center,
                  ),
                  const SizedBox(height: 12),
                  ElevatedButton(
                    onPressed: _muatUlang,
                    child: const Text('Coba lagi'),
                  ),
                ],
              ),
            ),
          );
        }

        final data = snapshot.data!;
        return ListView.builder(
          itemCount: data.length,
          itemBuilder: (context, i) {
            final p = data[i];
            return ListTile(
              leading: CircleAvatar(child: Text(p.name[0])),
              title: Text(p.name),
              subtitle: Text(p.email),
              trailing: const Icon(Icons.chevron_right),
              onTap: () {},
            );
          },
        );
      },
    ),
  );
 }
}
```

Hasil :


3:00
DEBUG
Daftar Pengguna


3:00
DEBUG
Daftar Pengguna
L
Leanne Graham
Sincere@april.biz
E
Ervin Howell
Shanna@melissa.tv
C
Clementine Bauch
Nathan@yesenia.net
P
Patricia Lebsack
Julianne.OConner@kory.org
C
Chelsey Dietrich
Lucio_Hettinger@annie.ca
M
Mrs. Dennis Schulist
Karley_Dach@jasper.info
K
Kurtis Weissnat
Telly.Hoeger@billy.biz
N
Nicholas Runolfsdottir V
Sherwood@rosamond.me
G
Glenna Reichert
Chaim_McDermott@dana.io
C
Clementina DuBuque
Rey.Padberg@karina.biz

Bagian E :

```dart
import 'dart:convert';
import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;

class Pengguna {
 final int id;
 final String name;
 final String email;
 final String phone;
 final String website;

 const Pengguna({
   required this.id,
   required this.name,
   required this.email,
   required this.phone,
   required this.website,
 });

 factory Pengguna.fromJson(Map<String, dynamic> json) {
   return Pengguna(
     id: json['id'] as int,
     name: json['name'] as String,
     email: json['email'] as String,
     phone: json['phone'] as String,
     website: json['website'] as String,
   );
 }
}

Future<List<Pengguna>> ambilPengguna() async {
 final uri = Uri.parse('https://jsonplaceholder.typicode.com/users');
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 return data
     .map((e) => Pengguna.fromJson(e as Map<String, dynamic>))
     .toList();
}

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
 const MyApp({super.key});

 @override
 Widget build(BuildContext context) {
   return MaterialApp(
```

```
      title: 'Praktikum 4',
      theme: ThemeData(
        colorSchemeSeed: Colors.blue,
        useMaterial3: true,
      ),
      home: const PenggunaPage(),
    );
  }
}

class PenggunaPage extends StatefulWidget {
  const PenggunaPage({super.key});

  @override
  State<PenggunaPage> createState() => _PenggunaPageState();
}

class _PenggunaPageState extends State<PenggunaPage> {
  late Future<List<Pengguna>> _future;

  @override
  void initState() {
    super.initState();
    _future = ambilPengguna();
  }

  void _muatUlang() {
    setState(() {
      _future = ambilPengguna();
    });
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Daftar Pengguna'),
        actions: [
          IconButton(
            icon: const Icon(Icons.refresh),
            onPressed: _muatUlang,
          ),
        ],
      ),
      body: FutureBuilder<List<Pengguna>>(
        future: _future,
        builder: (context, snapshot) {
          if (snapshot.connectionState == ConnectionState.waiting) {
            return const Center(child: CircularProgressIndicator());
          }
          if (snapshot.hasError) {
            return Center(
              child: Padding(
```

```
              padding: const EdgeInsets.all(24),
              child: Column(
                mainAxisSize: MainAxisSize.min,
                children: [
                  const Icon(
                    Icons.error_outline,
                    size: 48,
                    color: Colors.red,
                  ),
                  const SizedBox(height: 8),
                  Text(
                    'Terjadi kesalahan:\n${snapshot.error}',
                    textAlign: TextAlign.center,
                  ),
                  const SizedBox(height: 12),
                  ElevatedButton(
                    onPressed: _muatUlang,
                    child: const Text('Coba lagi'),
                  ),
                ],
              ),
            ),
          );
        }

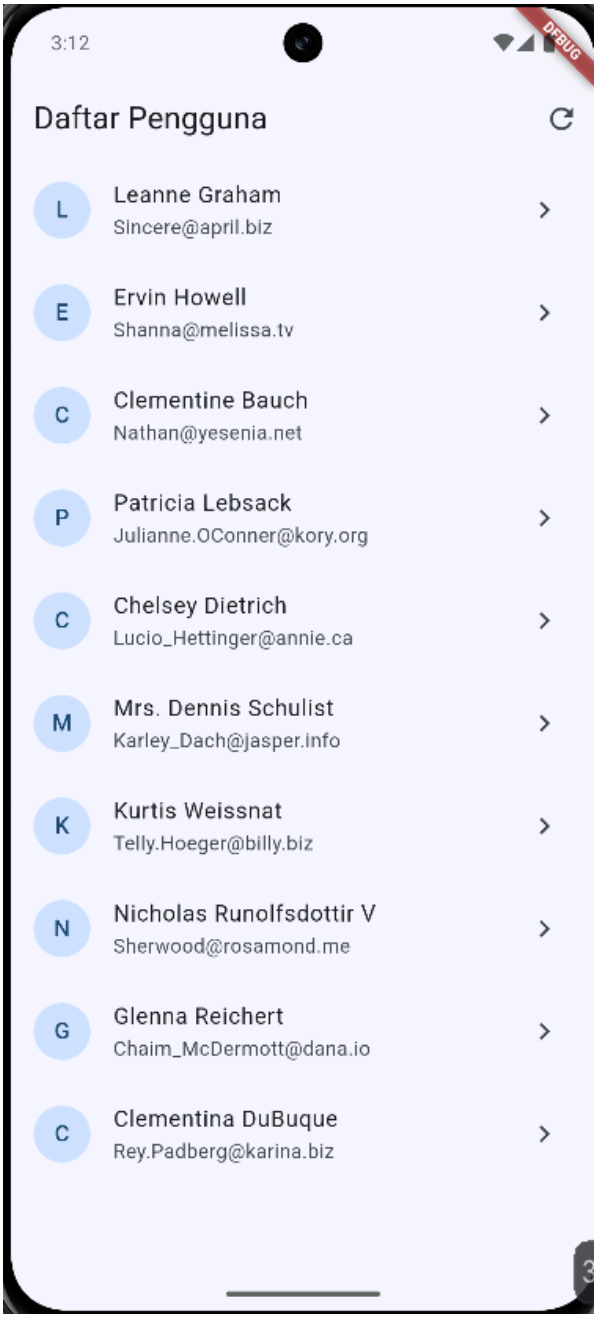
        final data = snapshot.data!;
        return ListView.builder(
          itemCount: data.length,
          itemBuilder: (context, i) {
            final p = data[i];
            return ListTile(
              leading: CircleAvatar(child: Text(p.name[0])),
              title: Text(p.name),
              subtitle: Text(p.email),
              trailing: const Icon(Icons.chevron_right),
              onTap: () {
                Navigator.push(
                  context,
                  MaterialPageRoute(
                    builder: (_) => DetailPenggunaPage(pengguna: p),
                  ),
                );
              },
            );
          },
        );
      },
    ),
  );
 }
}

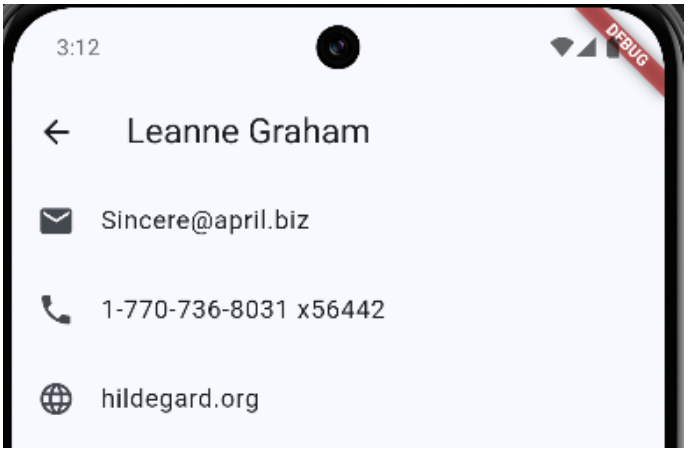
class DetailPenggunaPage extends StatelessWidget {
```

```dart
final Pengguna pengguna;
const DetailPenggunaPage({super.key, required this.pengguna});

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(title: Text(pengguna.name)),
    body: ListView(
      children: [
        ListTile(
          leading: const Icon(Icons.email),
          title: Text(pengguna.email),
        ),
        ListTile(
          leading: const Icon(Icons.phone),
          title: Text(pengguna.phone),
        ),
        ListTile(
          leading: const Icon(Icons.language),
          title: Text(pengguna.website),
        ),
      ],
    ),
  );
}
}
```

Hasil :


3:12
DEBUG
Daftar Pengguna
L
Leanne Graham
Sincere@april.biz
E
Ervin Howell
Shanna@melissa.tv
C
Clementine Bauch
Nathan@yesenia.net
P
Patricia Lebsack
Julianne.OConner@kory.org
C
Chelsey Dietrich
Lucio_Hettinger@annie.ca
M
Mrs. Dennis Schulist
Karley_Dach@jasper.info
K
Kurtis Weissnat
Telly.Hoeger@billy.biz
N
Nicholas Runolfsdottir V
Sherwood@rosamond.me
G
Glenna Reichert
Chaim_McDermott@dana.io
C
Clementina DuBuque
Rey.Padberg@karina.biz

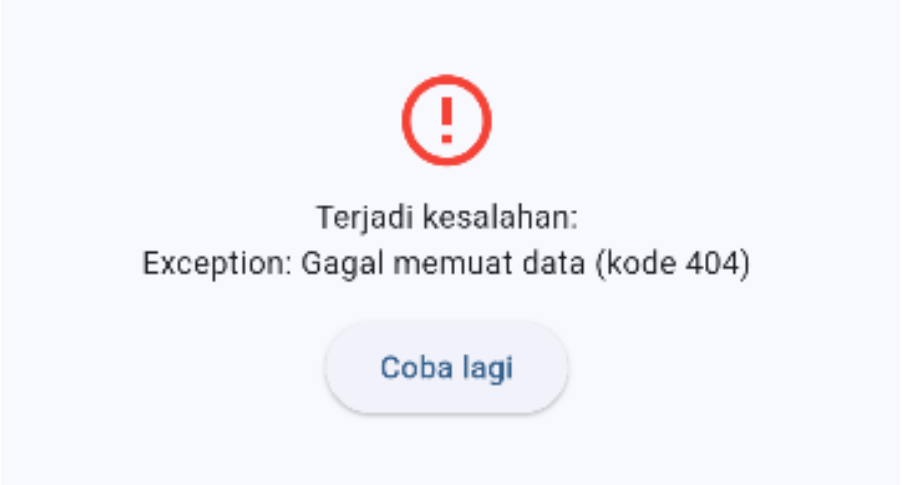

3:12
DEBUG
Leanne Graham
Sincere@april.biz
1-770-736-8031 x56442
hildegard.org

Bagian F :

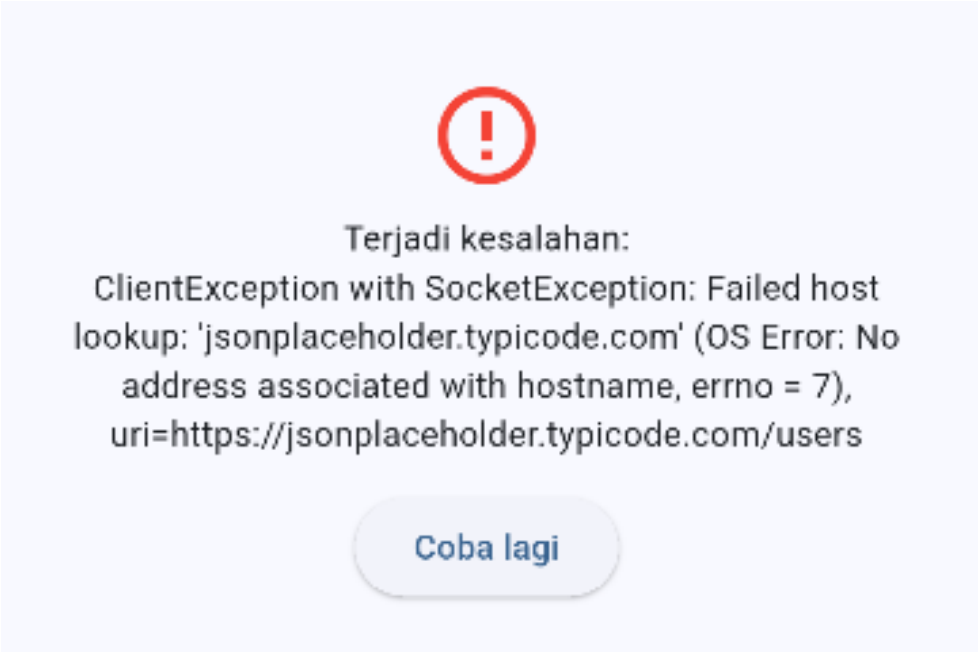
```
final uri = Uri.parse('https://jsonplaceholder.typicode.com/users');
```

Jadi :

```
final uri = Uri.parse('https://jsonplaceholder.typicode.com/usersz');
```



Jika wifi dimatikan :



Latihan Mandiri :

```dart
import 'dart:convert';
import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;

class Pengguna {
 final int id;
 final String name;
 final String username;
 final String email;
 final String phone;
 final String website;
 final String city;

 const Pengguna({
   required this.id,
   required this.name,
   required this.username,
   required this.email,
   required this.phone,
   required this.website,
   required this.city,
 });

 factory Pengguna.fromJson(Map<String, dynamic> json) {
   return Pengguna(
     id: json['id'] as int,
     name: json['name'] as String,
     username: json['username'] as String,
     email: json['email'] as String,
     phone: json['phone'] as String,
     website: json['website'] as String,
     city: json['address']['city'] as String,
   );
 }
}

Future<List<Pengguna>> ambilPengguna() async {
 final uri = Uri.parse('https://jsonplaceholder.typicode.com/users');
 final response = await http.get(uri).timeout(const Duration(seconds: 10));

 if (response.statusCode != 200) {
   throw Exception('Gagal memuat data (kode ${response.statusCode})');
 }

 final List<dynamic> data = jsonDecode(response.body);
 return data
     .map((e) => Pengguna.fromJson(e as Map<String, dynamic>))
     .toList();
}

void main() => runApp(const MyApp());
```

```dart
class MyApp extends StatelessWidget {
 const MyApp({super.key});

 @override
 Widget build(BuildContext context) {
   return MaterialApp(
     title: 'Praktikum 4',
     theme: ThemeData(
       colorSchemeSeed: Colors.blue,
       useMaterial3: true,
     ),
     home: const PenggunaPage(),
   );
 }
}

class PenggunaPage extends StatefulWidget {
 const PenggunaPage({super.key});

 @override
 State<PenggunaPage> createState() => _PenggunaPageState();
}

class _PenggunaPageState extends State<PenggunaPage> {
 late Future<List<Pengguna>> _future;

 @override
 void initState() {
   super.initState();
   _future = ambilPengguna();
 }

 Future<void> _muatUlang() async {
   setState(() {
     _future = ambilPengguna();
   });
   await _future;
 }

 @override
 Widget build(BuildContext context) {
   return Scaffold(
     appBar: AppBar(
       title: FutureBuilder<List<Pengguna>>(
         future: _future,
         builder: (context, snapshot) {
           if (snapshot.hasData) {
             return Text('Daftar Pengguna (${snapshot.data!.length})');
           }
           return const Text('Daftar Pengguna');
         },
       ),
       actions: [
```

```
        IconButton(
          icon: const Icon(Icons.refresh),
          onPressed: _muatUlang,
        ),
      ],
    ),
    body: FutureBuilder<List<Pengguna>>(
      future: _future,
      builder: (context, snapshot) {
        if (snapshot.connectionState == ConnectionState.waiting) {
          return const Center(child: CircularProgressIndicator());
        }
        if (snapshot.hasError) {
          return Center(
            child: Padding(
              padding: const EdgeInsets.all(24),
              child: Column(
                mainAxisSize: MainAxisSize.min,
                children: [
                  const Icon(
                    Icons.error_outline,
                    size: 48,
                    color: Colors.red,
                  ),
                  const SizedBox(height: 8),
                  Text(
                    'Terjadi kesalahan:\n${snapshot.error}',
                    textAlign: TextAlign.center,
                  ),
                  const SizedBox(height: 12),
                  ElevatedButton(
                    onPressed: _muatUlang,
                    child: const Text('Coba lagi'),
                  ),
                ],
              ),
            ),
          );
        }

        final data = snapshot.data!;

        if (data.isEmpty) {
          return RefreshIndicator(
            onRefresh: _muatUlang,
            child: ListView(
              physics: const AlwaysScrollableScrollPhysics(),
              children: const [
                SizedBox(height: 200),
                Center(child: Text('Tidak ada data')),
              ],
            ),
          );
```

```
        }

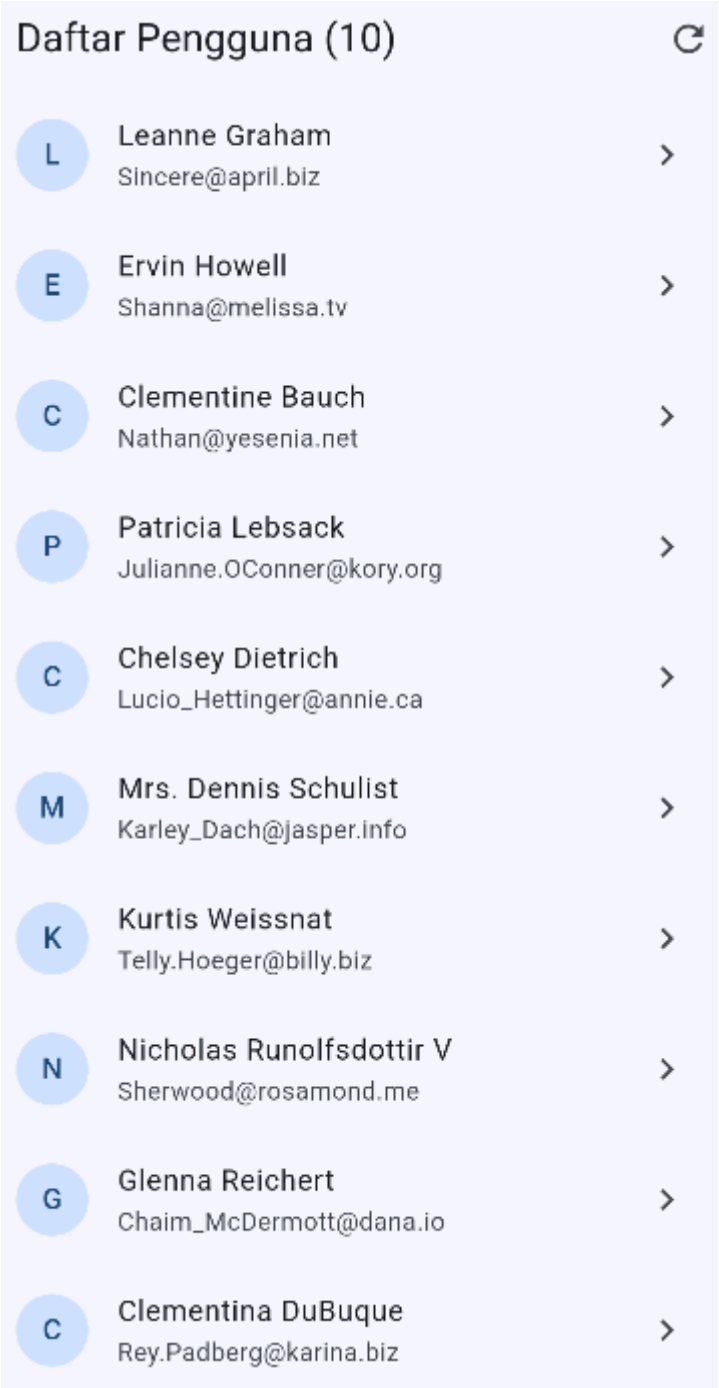
        return RefreshIndicator(
          onRefresh: _muatUlang,
          child: ListView.builder(
            itemCount: data.length,
            itemBuilder: (context, i) {
              final p = data[i];
              return ListTile(
                leading: CircleAvatar(child: Text(p.name[0])),
                title: Text(p.name),
                subtitle: Text(p.email),
                trailing: const Icon(Icons.chevron_right),
                onTap: () {
                  Navigator.push(
                    context,
                    MaterialPageRoute(
                      builder: (_) => DetailPenggunaPage(pengguna: p),
                    ),
                  );
                },
              );
            },
          ),
        );
      },
    ),
  );
 }
}

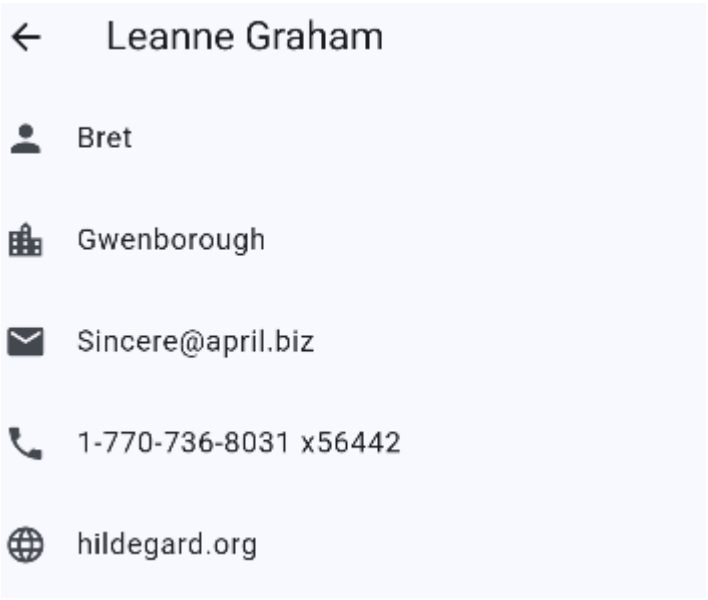
class DetailPenggunaPage extends StatelessWidget {
 final Pengguna pengguna;
 const DetailPenggunaPage({super.key, required this.pengguna});

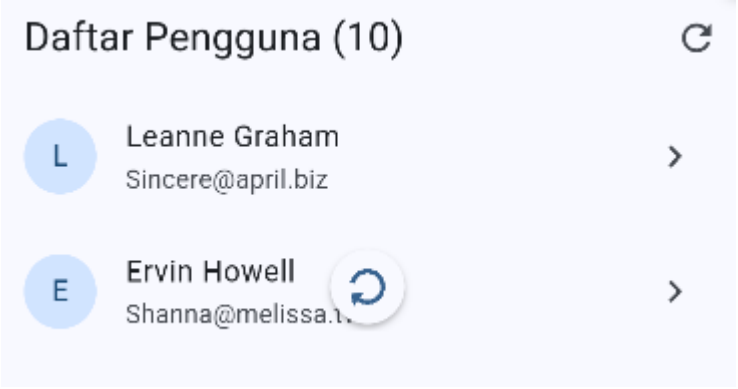
 @override
 Widget build(BuildContext context) {
   return Scaffold(
     appBar: AppBar(title: Text(pengguna.name)),
     body: ListView(
       children: [
         ListTile(
           leading: const Icon(Icons.person),
           title: Text(pengguna.username),
         ),
         ListTile(
           leading: const Icon(Icons.location_city),
           title: Text(pengguna.city),
         ),
         ListTile(
           leading: const Icon(Icons.email),
           title: Text(pengguna.email),
         ),
```

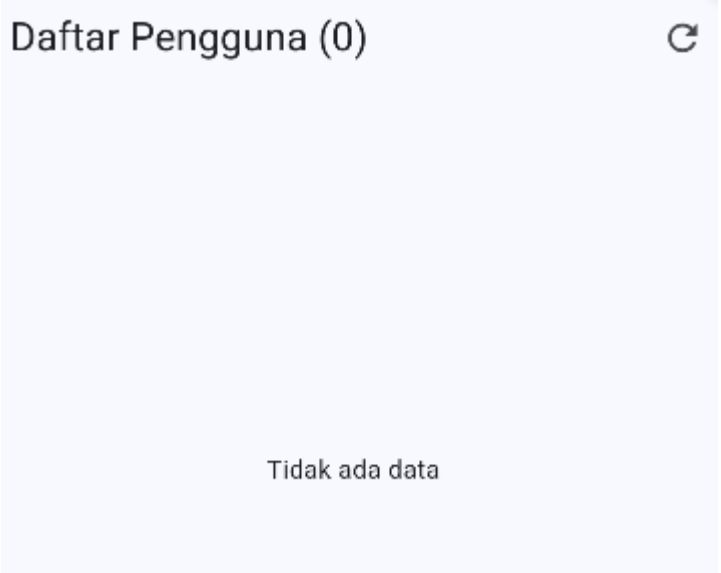
```
          ListTile(
            leading: const Icon(Icons.phone),
            title: Text(pengguna.phone),
          ),
          ListTile(
            leading: const Icon(Icons.language),
            title: Text(pengguna.website),
          ),
        ],
      ),
    );
  }
}
```

Hasil :


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Daftar Pengguna (0)
Tidak ada data

Tugas :

```dart
import 'dart:convert';
import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;

class Post {
 final int id;
 final String title;
 final String body;

 const Post({
   required this.id,
   required this.title,
   required this.body,
 });

 factory Post.fromJson(Map<String, dynamic> json) {
   return Post(
     id: json['id'] as int,
     title: json['title'] as String,
     body: json['body'] as String,
   );
 }
}

class Komentar {
 final int id;
 final String name;
 final String email;
 final String body;

 const Komentar({
   required this.id,
   required this.name,
   required this.email,
   required this.body,
 });

 factory Komentar.fromJson(Map<String, dynamic> json) {
   return Komentar(
     id: json['id'] as int,
     name: json['name'] as String,
     email: json['email'] as String,
     body: json['body'] as String,
   );
 }
}

Future<List<Post>> ambilPosts() async {
 final uri = Uri.parse('https://jsonplaceholder.typicode.com/posts');
 final response = await http.get(uri).timeout(const Duration(seconds: 10));

 if (response.statusCode != 200) {
```

```
    throw Exception('Gagal memuat postingan (kode ${response.statusCode})');
  }

  final List<dynamic> data = jsonDecode(response.body);
  return data.map((e) => Post.fromJson(e as Map<String, dynamic>)).toList();
}

Future<List<Komentar>> ambilKomentar(int postId) async {
  final uri =
Uri.parse('https://jsonplaceholder.typicode.com/posts/$postId/comments');
  final response = await http.get(uri).timeout(const Duration(seconds: 10));

  if (response.statusCode != 200) {
    throw Exception('Gagal memuat komentar (kode ${response.statusCode})');
  }

  final List<dynamic> data = jsonDecode(response.body);
  return data.map((e) => Komentar.fromJson(e as Map<String,
dynamic>)).toList();
}

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Aplikasi Postingan',
      theme: ThemeData(
        colorSchemeSeed: Colors.blue,
        useMaterial3: true,
      ),
      home: const PostPage(),
    );
  }
}

class PostPage extends StatefulWidget {
  const PostPage({super.key});

  @override
  State<PostPage> createState() => _PostPageState();
}

class _PostPageState extends State<PostPage> {
  late Future<List<Post>> _future;

  @override
  void initState() {
    super.initState();
    _future = ambilPosts();
```

```dart
}

Future<void> _muatUlang() async {
  setState(() {
    _future = ambilPosts();
  });
  await _future;
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text('Daftar Postingan'),
      actions: [
        IconButton(
          icon: const Icon(Icons.refresh),
          onPressed: _muatUlang,
        ),
      ],
    ),
    body: FutureBuilder<List<Post>>(
      future: _future,
      builder: (context, snapshot) {
        if (snapshot.connectionState == ConnectionState.waiting) {
          return const Center(child: CircularProgressIndicator());
        }
        if (snapshot.hasError) {
          return Center(
            child: Padding(
              padding: const EdgeInsets.all(24),
              child: Column(
                mainAxisSize: MainAxisSize.min,
                children: [
                  const Icon(Icons.error_outline, size: 48, color:
Colors.red),
                  const SizedBox(height: 8),
                  Text(
                    'Terjadi kesalahan:\n${snapshot.error}',
                    textAlign: TextAlign.center,
                  ),
                  const SizedBox(height: 12),
                  ElevatedButton(
                    onPressed: _muatUlang,
                    child: const Text('Coba lagi'),
                  ),
                ],
              ),
            ),
          );
        }

        final data = snapshot.data!;
```

```
        return RefreshIndicator(
          onRefresh: _muatUlang,
          child: ListView.builder(
            itemCount: data.length,
            itemBuilder: (context, i) {
              final post = data[i];
              return ListTile(
                title: Text(
                  post.title,
                  maxLines: 1,
                  overflow: TextOverflow.ellipsis,
                ),
                subtitle: Text(
                  post.body,
                  maxLines: 2,
                  overflow: TextOverflow.ellipsis,
                ),
                trailing: const Icon(Icons.chevron_right),
                onTap: () {
                  Navigator.push(
                    context,
                    MaterialPageRoute(
                      builder: (_) => DetailPostPage(post: post),
                    ),
                  );
                },
              );
            },
          ),
        );
      },
    ),
  );
 }
}

class DetailPostPage extends StatefulWidget {
 final Post post;
 const DetailPostPage({super.key, required this.post});

 @override
 State<DetailPostPage> createState() => _DetailPostPageState();
}

class _DetailPostPageState extends State<DetailPostPage> {
 late Future<List<Komentar>> _futureKomentar;

 @override
 void initState() {
   super.initState();
   _futureKomentar = ambilKomentar(widget.post.id);
 }
```

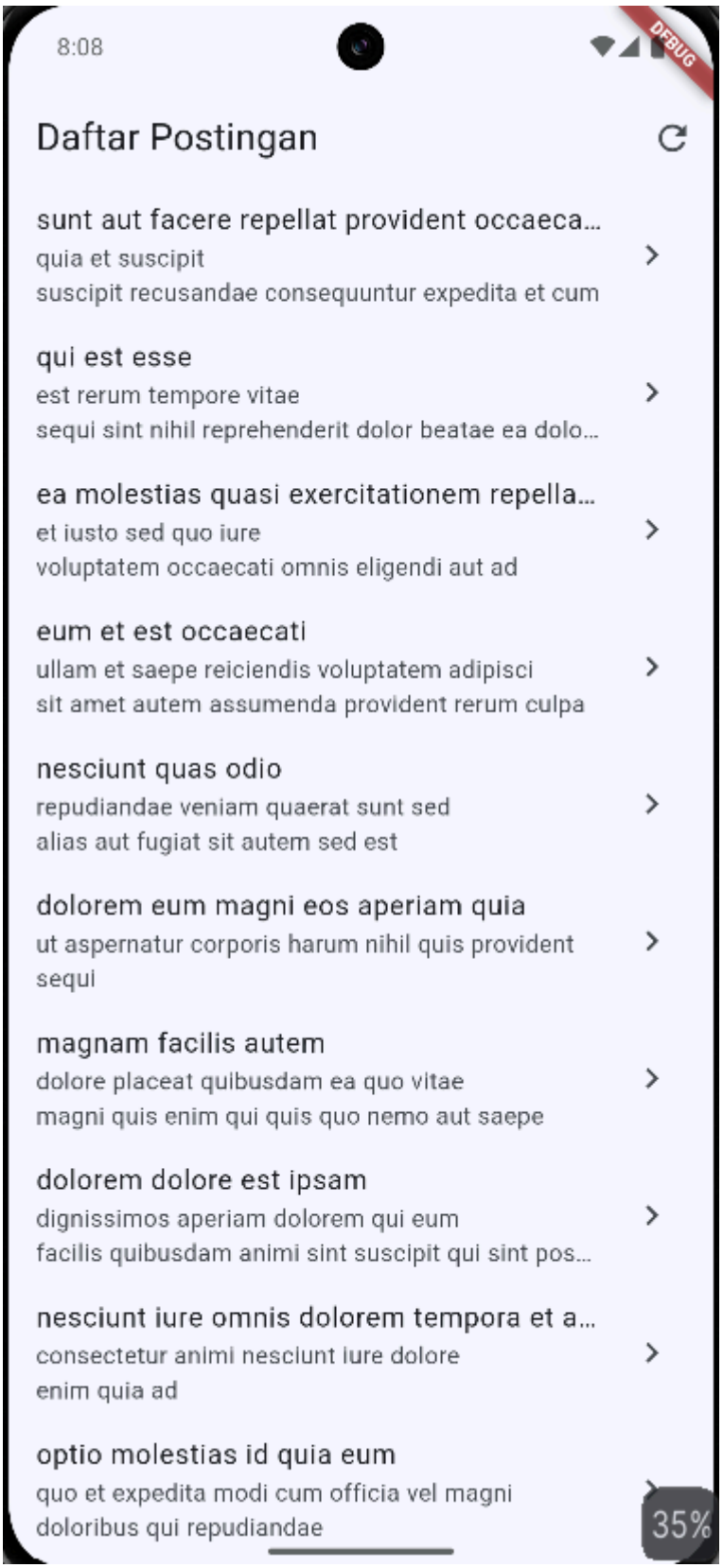
```
 void _muatUlangKomentar() {
   setState(() {
     _futureKomentar = ambilKomentar(widget.post.id);
   });
 }

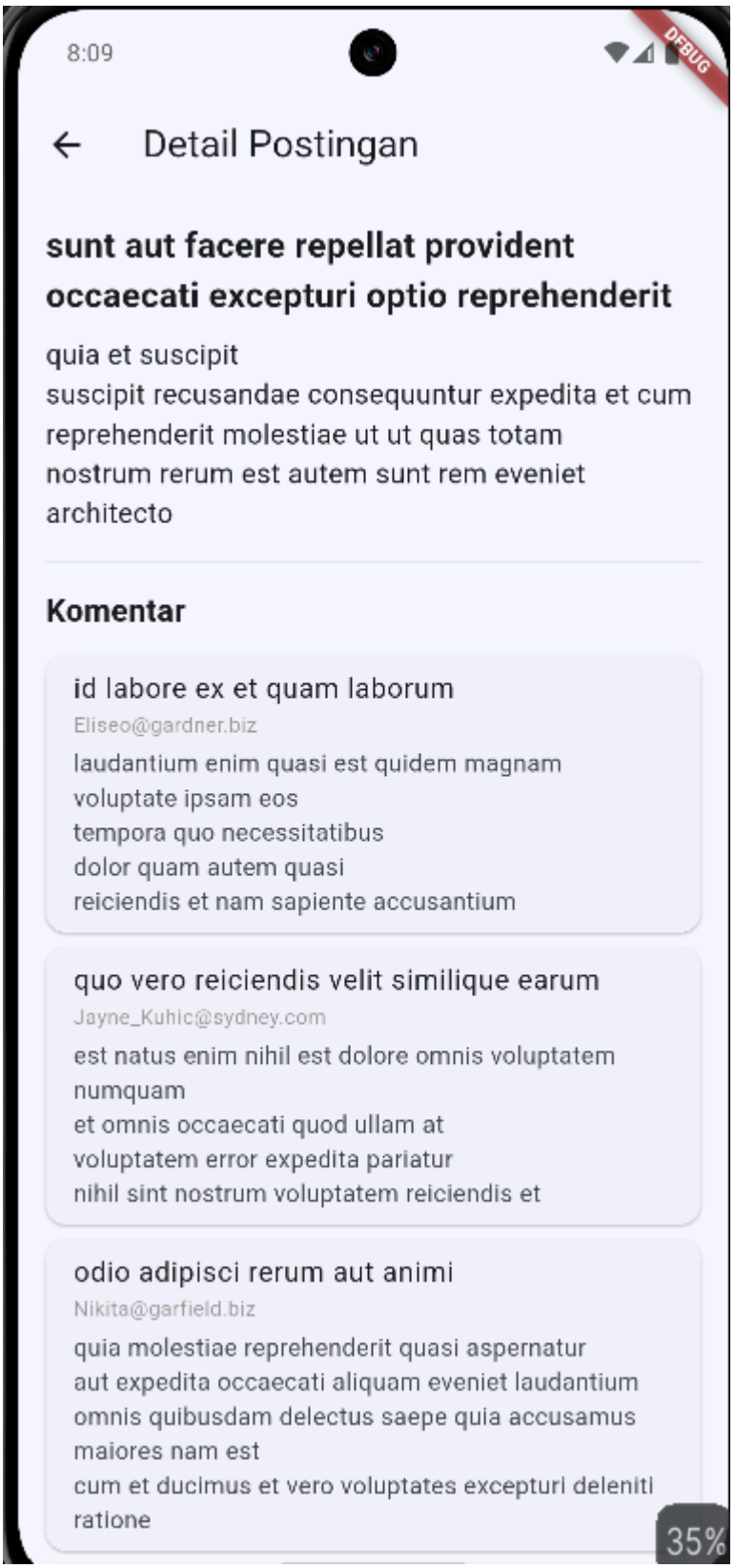
 @override
 Widget build(BuildContext context) {
   return Scaffold(
     appBar: AppBar(
       title: const Text('Detail Postingan'),
     ),
     body: SingleChildScrollView(
       padding: const EdgeInsets.all(16),
       child: Column(
         crossAxisAlignment: CrossAxisAlignment.start,
         children: [
           Text(
             widget.post.title,
             style: const TextStyle(fontSize: 20, fontWeight:
FontWeight.bold),
           ),
           const SizedBox(height: 8),
           Text(
             widget.post.body,
             style: const TextStyle(fontSize: 16),
           ),
           const Divider(height: 32),
           const Text(
             'Komentar',
             style: TextStyle(fontSize: 18, fontWeight: FontWeight.bold),
           ),
           const SizedBox(height: 8),
           FutureBuilder<List<Komentar>>(
             future: _futureKomentar,
             builder: (context, snapshot) {
               if (snapshot.connectionState == ConnectionState.waiting) {
                 return const Center(
                   child: Padding(
                     padding: EdgeInsets.all(16),
                     child: CircularProgressIndicator(),
                   ),
                 );
               }
               if (snapshot.hasError) {
                 return Center(
                   child: Column(
                     children: [
                       Text('Gagal memuat komentar: ${snapshot.error}'),
                       const SizedBox(height: 8),
                       ElevatedButton(
                         onPressed: _muatUlangKomentar,
                         child: const Text('Coba lagi'),
```

```
                    ),
                  ],
                ),
              );
            }

            final komentarList = snapshot.data!;
            return ListView.builder(
              shrinkWrap: true,
              physics: const NeverScrollableScrollPhysics(),
              itemCount: komentarList.length,
              itemBuilder: (context, i) {
                final k = komentarList[i];
                return Card(
                  margin: const EdgeInsets.symmetric(vertical: 4),
                  child: ListTile(
                    title: Text(k.name),
                    subtitle: Column(
                      crossAxisAlignment: CrossAxisAlignment.start,
                      children: [
                        Text(
                          k.email,
                          style: const TextStyle(
                            fontSize: 12,
                            color: Colors.grey,
                          ),
                        ),
                        const SizedBox(height: 4),
                        Text(k.body),
                      ],
                    ),
                  ),
                );
              },
            );
          },
        ),
      ],
    ),
  ),
 );
}
}
```

Hasil :


8:08
DEBUG
Daftar Postingan
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suscipit recusandae consequuntur expedita et cum
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est rerum tempore vitae
sequi sint nihil reprehenderit dolor beatae ea dolo...
ea molestias quasi exercitationem repella...
et iusto sed quo iure
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eum et est occaecati
ullam et saepe reiciendis voluptatem adipisci
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ut aspernatur corporis harum nihil quis provident
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8:09
DEBUG
Detail Postingan
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architecto
Komentar
id labore ex et quam laborum
Eliseo@gardner.biz
laudantium enim quasi est quidem magnam
voluptate ipsam eos
tempora quo necessitatibus
dolor quam autem quasi
reiciendis et nam sapiente accusantium
quo vero reiciendis velit similique earum
Jayne_Kuhic@sydney.com
est natus enim nihil est dolore omnis voluptatem
numquam
et omnis occaecati quod ullam at
voluptatem error expedita pariatur
nihil sint nostrum voluptatem reiciendis et
odio adipisci rerum aut animi
Nikita@garfield.biz
quia molestiae reprehenderit quasi aspernatur
aut expedita occaecati aliquam eveniet laudantium
omnis quibusdam delectus saepe quia accusamus
maiores nam est
cum et ducimus et vero voluptates excepturi deleniti
ratione
35%



Terjadi kesalahan:
ClientException with SocketException: Failed host lookup: 'jsonplaceholder.typicode.com' (OS Error: No address associated with hostname, errno = 7), uri=https://jsonplaceholder.typicode.com/posts

Coba lagi